



Answers and Solutions

1. B. 4.50 s

Use the formula for total time of flight of a projectile motion.

$$T = \frac{2v_o \sin \theta}{g} = \frac{2(44) \sin 30^\circ}{9.81}$$

$$T = 4.49 \text{ s} \rightarrow \text{Ans}$$

Reference: GEAS 04-12 – Projectile Motion 1

2. C. 11.4 N

$$F \cos 30^\circ = 0.05(20)(9.81)$$

$$F = 11.4 \text{ N}$$

Reference: GEAS 04-01 – Types of Equilibrium

3. D. 30 m, downwards

Displacement is the change in position:

$$\Delta x = x_f - x_o = 0 - 30 = -30 \text{ m}$$

Reference: GEAS 04-08 – Vertical Motion

4. C. 27 m

$$v_f = v_o + at$$

$$14 = 4 + a(3)$$

$$a = 3.33 \frac{\text{m}}{\text{s}^2}$$

$$s = v_o t + \frac{1}{2} at^2$$

$$s = \left(4 \frac{\text{m}}{\text{s}}\right)(3 \text{ s}) + \frac{1}{2} \left(3.33 \frac{\text{m}}{\text{s}^2}\right)(3 \text{ s})^2$$

$$s = 27 \text{ m}$$

Reference: GEAS 04-07 – Horizontal Motion

5. C. 56 m

$$s = v_o t + \frac{1}{2} at^2$$

$$s = \left(10 \frac{\text{m}}{\text{s}}\right)(4 \text{ s}) + \frac{1}{2} \left(2 \frac{\text{m}}{\text{s}^2}\right)(4 \text{ s})^2$$

$$s = 56 \text{ m}$$

Reference: GEAS 04-07 – Horizontal Motion

6. B. Velocity

The slope of a position vs time graph gives you the velocity.

Reference: GEAS 04-09 – Graphical Analysis

7. B. 10 m/s²

The rock is a free-falling object, so the acceleration is closest to g.

Reference: GEAS 04-08 – Vertical Motion

8. D. acceleration

The slope of a velocity vs time graph gives you the acceleration.

Reference: GEAS 04-09 – Graphical Analysis

9. B. 4 s

$$s = v_o t + \frac{1}{2} at^2$$

$$0 - 80 = 0 - \frac{1}{2} \left(9.81 \frac{\text{m}}{\text{s}^2}\right) t^2$$

$$t = 4 \text{ s}$$

Reference: GEAS 04-08 – Vertical Motion

10. D. 40 m/s

$$v_f = v_o + at$$

$$v_f = -20 - (9.81)(2) = -40 \frac{\text{m}}{\text{s}}$$

Reference: GEAS 04-08 – Vertical Motion

11. C. 21 m

Use the formula of displacement:

$$s = v_o t + \frac{1}{2} at^2$$

$$s = \left(8 \frac{\text{m}}{\text{s}}\right)(3 \text{ s}) - \frac{1}{2} \left(9.81 \frac{\text{m}}{\text{s}^2}\right)(3 \text{ s})^2$$

$$s = -21 \text{ m}$$

This means that the displacement in 3 s is 21 m downwards OR it was launched from a height 21 m above the ground.

Reference: GEAS 04-08 – Vertical Analysis

12. C. 30 m

Since the horizontal component of the projectile motion is not accelerating:

$$s_x = v_{ox} t + \frac{1}{2} a_x t^2$$

$$s_x = (15)(2) = 30 \text{ m}$$

Reference: GEAS 04-13 – Projectile Motion 2

13. C. The horizontal velocity did not change

In projectile motion, the horizontal component of the velocity will never change. Only the vertical component of the velocity is changing.

Reference: GEAS 04-13 – Projectile Motion 1

14. D. 6 s

Use the formula for the total time of flight:

$$T = \frac{2v_o \sin \theta}{g} = \frac{2(60) \sin 30^\circ}{9.81}$$

$$T = 6.12 \text{ s} \rightarrow \text{Ans}$$

Reference: GEAS 04-12 – Projectile Motion 1

15. A. I only

II is false – average speed of uniform circular motion is not 0

III is false – an object circular motion will ALWAYS acceleration since the direction of the velocity vector keeps on changing.

Reference: GEAS 04-14 – Uniform Circular Motion

16. B. II only



I is false – an object can be accelerating while not moving in a straight line (e.g. uniform circular motion)

III is false – if speed is constant, acceleration is not necessarily zero (e.g. uniform circular motion)

Reference: GEAS 04-14 – Uniform Circular Motion

17. C. g, downwards

At the highest point of the vertical motion, the acceleration of a free-falling object is due to gravity g.

Reference: GEAS 04-08 – Vertical Motion

18. D. vertical speed decreases on the way up and increases on the way down

For projectile motion, since the acceleration is due to gravity, as the object goes up it's slowing down vertically and as it goes down it's moving faster and faster. However, the horizontal component of the velocity will not change because there is no horizontal acceleration.

Reference: GEAS 04-12 – Projectile Motion 1

19. C. 50 m/s

Analyze the horizontal component of this projectile motion:

$$v_{fx} = v_{ox} + a_x t$$

$$v_{fx} = v_{ox} + a_x t$$

$$v_{fx} = 30 \frac{m}{s}$$

Analyze the vertical component of this projectile motion:

$$v_{fy} = v_{oy} \pm a_y t$$

$$v_{fy} = v_{oy} - gt$$

$$v_{fy} = 0 - 9.81(4) = -39.24 \frac{m}{s}$$

Get the total velocity after 4 s:

$$v = \sqrt{30^2 + (-39.24)^2} = 50 \frac{m}{s}$$

Reference: GEAS 04-13 – Projectile Motion 2

20. A. 0

“constant speed” means acceleration is 0.

Reference: GEAS 04-07 – Horizontal Motion

21. C. 4 m/s²

$$a = \frac{\Delta v}{t} = \frac{32 - 20}{3} = 4 \frac{m}{s^2}$$

Reference: GEAS 04-07 – Horizontal Motion

22. A. displacement

The area between the graph and the time axis of a velocity vs time graph will give you the displacement.

Reference: GEAS 04-09 – Graphical Analysis

23. D. The crate will not move.

Solve for the maximum static friction:

$$f_{s(max)} = \mu_s N = (0.4) \left[(100 \text{ kg}) \left(9.81 \frac{m}{s^2} \right) \right]$$

$$f_{s(max)} = 392.4 \text{ N}$$

Since the force applied did not exceed the maximum static friction, the box will not yet move.

24. A. force of the floor on crate #1 is greater

If you push only 1 box:

$$\Sigma F_y = 0$$

$$N - W = 0$$

$$N = W$$

$$\Sigma F_x = 0$$

$$P - f = 0$$

$$P = f = \mu N = \mu mg$$

If you push only 2 boxes with the same mass (one on top of the other):

$$\Sigma F_y = 0$$

$$N - 2W = 0$$

$$N = 2W = 2mg$$

$$\Sigma F_x = 0$$

$$P' - f = 0 = f = \mu N = \mu 2mg = 2P$$

P' is greater than P because the normal force of the floor on the boxes is greater in the second scenario.

25. A. accelerate down the ramp

Compare the component of the weight along the x-axis and the value of the maximum static friction.

$$W_x = mg \sin 30^\circ$$

$$f_{s(max)} = \mu_s N = 0.5mg \cos 30^\circ$$

Since W_x is greater than $f_{s(max)}$, the box will accelerate down the ramp.

Reference: GEAS 04-10 – Inclined Plane Motion

26. A. 1.5 m/s²

$$\Sigma F_x = ma$$

$$90 - 0.3 \left[(20) (9.81) \right] = 20a$$

$$a = 1.6 \frac{m}{s^2}$$

Reference: GEAS 04-07 – Horizontal Motion

27. B. 250 N

Solve for the maximum static friction:

$$f_{s(max)} = \mu_s N = (0.40) \left[(100 \text{ kg}) \left(9.81 \frac{m}{s^2} \right) \right]$$

$$f_{s(max)} = 392.4 \text{ N}$$

Since the force applied is just 250 N (less than the maximum static friction), then the static friction present is just 250 N.

Reference: GEAS 04-07 – Horizontal Motion



28. C. 90 N

$$F_c = \frac{mv^2}{r} = \frac{(5)(6)^2}{2} = 90\text{N}$$

Reference: GEAS 04-14 – Uniform Circular Motion

29. B. 12 m/s

$$F = \frac{mv^2}{r}$$

$$1500\text{N} = \frac{(10\text{kg})(v)^2}{0.90\text{m}}$$

$$v = 12 \frac{\text{m}}{\text{s}}$$

Reference: GEAS 04-14 – Uniform Circular Motion

30. B. – 80 m

Analyze the vertical component of this projectile motion:

$$y = v_{oy}t + \frac{1}{2}a_yt^2 = 0 - \frac{1}{2}(9.81)(4)^2$$

$$y = -80\text{m}$$

Reference: GEAS 04-13 – Projectile Motion 2

31. A. $g(\sin q - \mu \cos q)$

Solution:

Since the block will accelerate down the plane, take the summation of forces in y-axis (perpendicular to the plane) and equate to zero while the summation of forces in both x-axis (parallel to the plane) is equal to ma . Take the axis parallel to the plane as the x-axis and the axis perpendicular to the plane as the y-axis. Draw your FBD. From the FBD:

$$\sum F_y = 0$$

$$N = W_y = mg \cos q$$

$$\sum F_x = \pm ma$$

$$W_x - f = ma$$

$$ma = mg \sin q - \mu mg \cos q$$

$$a = g \sin q - \mu g \cos q$$

$$a = g(\sin q - \mu \cos q) \rightarrow \text{Ans}$$

Reference: GEAS 04-10 – Inclined Plane Motion

32. A. 4.9 m/s²

$$\sum F_x = ma$$

$$mg \sin 30^\circ = ma$$

$$a = 4.91 \frac{\text{m}}{\text{s}^2}$$

Reference: GEAS 04-10 – Inclined Plane Motion

33. A. 2.5 rad/s²

$$w_f^2 = w_o^2 + 2aq$$

$$10^2 = 0^2 + 2a(20)$$

$$a = 2.5 \frac{\text{rad}}{\text{s}^2}$$

Reference: GEAS 04-15 – Non-Uniform Circular Motion

34. B. 9 rad

$$w_f = w_o + at \quad w_f^2 = w_o^2 + 2aq$$

$$5 = 1 + a(3) \quad 5^2 = 1^2 + 2(1.33)q$$

$$a = 1.33 \frac{\text{rad}}{\text{s}^2} \quad q = 9 \text{ rad}$$

Reference: GEAS 04-15 – Non-Uniform Circular Motion

35. D. rw^2

$$a_n = \frac{v^2}{r} = \frac{(rw)^2}{r} = rw^2$$

Reference: GEAS 04-14 – Uniform Circular Motion

36. A. 3 m/s²

$$w_f = w_o + at$$

$$5 = 2 + a(0.5)$$

$$a = 6 \frac{\text{rad}}{\text{s}^2}$$

$$a = ra = (0.5\text{m})\left(6 \frac{\text{rad}}{\text{s}^2}\right)$$

$$a = 3 \frac{\text{m}}{\text{s}^2}$$

Reference: GEAS 04-15 – Non-Uniform Circular Motion

37. A. 1.2 m/s

$$v = rw$$

$$v = (0.20\text{m})\left(6 \frac{\text{rad}}{\text{s}}\right) = 1.2 \frac{\text{m}}{\text{s}}$$

Reference: GEAS 04-14 – Uniform Circular Motion

38. C. 6 rad/s²

$$w_f = w_o + at$$

$$5 = 2 + a(0.5)$$

$$a = 6 \frac{\text{rad}}{\text{s}^2}$$

Reference: GEAS 04-15 – Non-Uniform Circular Motion

39. C. pi

$$w = \frac{Dq}{Dt} = \frac{2p \text{ rad}}{2 \text{ s}} = p \frac{\text{rad}}{\text{s}}$$

Reference: GEAS 04-14 – Uniform Circular Motion

40. D. 4 s

“at the origin” means $x = 0$

$$x(t) = 4t^2 - 6t - 40$$

$$0 = 4t^2 - 6t - 40$$

$$t = 4 \text{ s}$$

$$t = -\frac{5}{2} \text{ s}$$

Consider only the positive value for time.

Reference: GEAS 04-07 – Horizontal Motion

41. B. 26 m/s

Based from the previous problem, the object is at the origin at $t = 4 \text{ s}$. Use the definition of instantaneous velocity:



$$v = \frac{dx}{dt} = 8t - 6$$

$$v = 8(4) - 6$$

$$v = 26 \frac{m}{s}$$

Reference: GEAS 04-07 – Horizontal Motion

42. B. 25.4 ft/s

$$\sin q = \frac{4}{30}$$

$$q = 7.66^\circ$$

$$\tan q = \frac{v^2}{Rg}$$

$$\tan 7.66^\circ = \frac{v^2}{(150)(32.2)}$$

$$v = 25.4 \text{ ft/s} \rightarrow \text{Ans}$$

Reference: GEAS 04-17 – Banking of Curves

43. A. 85 min

$$R = R_{\text{earth}} + 100$$

$$R = 6.4 \times 10^6 \text{ m} + 1.6 \times 10^5 \text{ m}$$

$$R = 6.5 \times 10^6 \text{ m}$$

$$F_{\text{gravity}} = F_c$$

$$mg = \frac{mv^2}{R}$$

$$g = \frac{v^2}{R}$$

$$v = \sqrt{\left(9.81 \frac{m}{s^2}\right)(6.5 \times 10^6 \text{ m})}$$

$$v = 8000 \frac{m}{s}$$

$$T = \frac{2\pi R}{v} = \frac{2\pi(6.5 \times 10^6 \text{ m})}{8000 \frac{m}{s}}$$

$$T = 85 \text{ min} \rightarrow \text{Ans}$$

Reference: GEAS 04-14 – Uniform Circular Motion

44. D. 69.6 deg

$$\tan q = \frac{v^2}{Rg}$$

$$\tan q = \frac{\left(215 \frac{m}{s}\right)^2}{(1750 \text{ m})\left(9.81 \frac{m}{s^2}\right)}$$

$$q = 69.6^\circ \rightarrow \text{Ans}$$

Reference: GEAS 04-17 – Banking of Curves

45. C. 0.42 rad/s²

$$a = \frac{w_f - w_i}{t}$$

$$a = \frac{\left[4 \frac{\text{rev}}{\text{min}} \times \frac{2\pi \text{ rad}}{1 \text{ rev}} \times \frac{1 \text{ min}}{60 \text{ s}}\right] - 0}{10 \text{ s}}$$

$$a = 0.42 \text{ rad/s}^2 \rightarrow \text{Ans}$$

Reference: GEAS 04-15 – Non-uniform Circular Motion

46. C. 54 m/s

$$v = x^3 - 2x^2 - 5x + 4$$

$$v = (5)^3 - 2(5)^2 - 5(5) + 4$$

$$v = 54 \text{ m/s} \rightarrow \text{Ans}$$

Reference: GEAS 04-07 – Horizontal Motion

47. A. 21.7 m/s

Note: since the curve is “unbanked”, angle of elevation $\theta = 0$; Φ is called angle of friction and is equal to $\text{Arctan } \mu$ where μ is the coefficient of friction

$$\tan(q + f) = \frac{(v_{\text{max}})^2}{Rg}$$

$$\tan(0 + \tan^{-1} 0.80) = \frac{(v_{\text{max}})^2}{(60)(9.81)}$$

$$v_{\text{max}} = 21.7 \text{ m/s} \rightarrow \text{Ans}$$

Reference: GEAS 04-17 – Banking of Curves

48. A. 32.2 s

$$h = \frac{1}{2}gt_1t_2$$

$$h = \frac{1}{2}(32.2)(1)(2) = 32.2 \text{ s} \rightarrow \text{Ans}$$

Reference: GEAS 04-08 – Vertical Motion

49. C. 60 ft

$$v = v_o \pm at$$

$$120 = 0 + a(4); a = 30 \text{ ft/s}^2$$

$$s = v_o t \pm \frac{1}{2}at^2$$

$$s = 0 + \frac{1}{2}(30 \text{ ft/s}^2)(2)^2$$

$$s = 60 \text{ ft} \rightarrow \text{Ans}$$

Reference: GEAS 04-07 – Horizontal Motion

50. C. 15.23 deg

Vertical component of the motion:

$$y = V_{oy}t \pm \frac{1}{2}a_yt^2$$

$$-2400 = 0 - \frac{1}{2}(9.81)t^2$$

$$t = 22.12 \text{ s} \rightarrow \text{Ans}$$

Horizontal component of the motion:

$$x = V_{ox}t \pm \frac{1}{2}a_xt^2$$

$$x = (1200)\frac{1000}{3600}(22.12) + \frac{1}{2}(0.6)(9.81)(22.12)^2$$

$$x = 8813 \text{ m}$$

$$\tan q = \frac{2400}{8813}$$

$$q = 15.23^\circ \rightarrow \text{Ans}$$

Reference: GEAS 04-13 – Projectile Motion 2